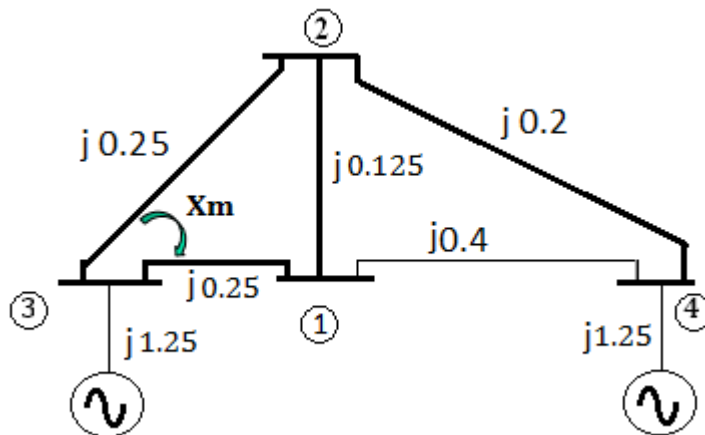


GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021****Subject Code:3160920****Date:04/12/2021****Subject Name:Inter Connected Power System****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- | | Marks |
|--|-----------|
| Q.1 (a) What is power system islanding? Why it is needed? | 03 |
| (b) Derive the Static power flow equations and also discuss the types of buses. | 04 |
| (c) Formulate Y_{BUS} using singular transformation Method for given network (line 1-3 and 2-3 are mutually coupled and $X_m = j 0.2$) | 07 |



- | | |
|---|-----------|
| Q.2 (a) What do you mean by Infinite bus with reference to power system? | 03 |
| (b) Write a short note on load dispatch Centre. | 04 |
| (c) Explain the Fast Decoupled load flow method with necessary assumptions | 07 |
| OR | |
| (c) Discuss the Z_{bus} building algorithm. | 07 |
| Q.3 (a) Discuss the steady state stability defining synchronizing power coefficient. | 03 |
| (b) Discuss the equal area criterion considering specific case in brief. | 04 |
| (c) | 07 |
- | Bus to Bus | 1-2 | 1-3 | 2-3 | 2-4 | 3-4 |
|------------|-----|-----|-------|-----|-----|
| G(pu) | 2 | 1 | 1.5 | 1 | 2 |
| B(pu) | -j8 | -j4 | -j6.5 | -j4 | -j8 |

Bus & Type	P_{gen}	Q_{gen}	P_{dem}	Q_{dem}	Bus Voltage
1 Slack	?	?	0	0	1.04
2 PQ	0	0	0.5	0.2	?
3 PQ	0	0	0.4	0.3	?
4 PQ	0	0	0.3	0.1	?

For a given power system data obtain voltages at bus no 2, 3 and 4 using Gauss – Seidal method

OR

- Q.3** (a) What do you mean by primitive network? **03**
 (b) Derive the equation describing the rotor dynamics. **04**
 (c) A 200 MVA, 11kV, 50 Hz, 4 pole turbo generator has an inertia constant of 6 - MJ/MVA. 1) Find the stored energy in the rotor at synchronous speed 2) machine operating at load of 120 MW and the load suddenly increases to 160 MW. Find the rotor retardation Neglecting losses. 3) If the retardation is maintained for 5 cycles, find the change in power angle and rotor speed in rpm at the end of the period. **07**

- Q.4** (a) Define critical clearing angle and critical clearing time **03**
 (b) Compare GS and NR method of load flow. **04**
 (c) Discuss the procedure of solving swing equation using point by point method. **07**

OR

- Q.4** (a) What you mean by cascade tripping **03**
 (b) Write short note on unit commitment. **04**
 (c) With the help of a neat diagram explain turbine speed governing system. **07**
- Q.5** (a) Describe the methods of voltage control adopted for large size power system in brief. **03**
 (b) Explain the terms 1) Incremental fuel cost 2) Penalty factor **04**
 (c) Incremental fuel costs in dollars per megawatthour for a plant consisting of two units are given by **07**

$$\lambda_1 = \frac{dC_1}{dP_1} = 0.008 * P_1 + 8 \quad , \quad \lambda_2 = \frac{dC_2}{dP_2} = 0.0096 * P_2 + 6.4$$

Assume that both units are operating at all times, that total load varies from 250 MW to 1250 MW, and that maximum and minimum loads on each unit are to be 625 and 100 MW , respectively. Fins the Incremental fuel cost of the plant and the allocation of load between units for the minimum cost of various total loads.

OR

- Q.5** (a) What is cascade tripping? **03**
 (b) Explain tie line load bias method of frequency control in brief. **04**
 (c) Derive the expression for B-coefficients in case of two generating plants connected to an arbitrary number of loads through a transmission network. **07**
